STDISCM Problem Set 3
Parallel OCR Pipeline [80 pts]

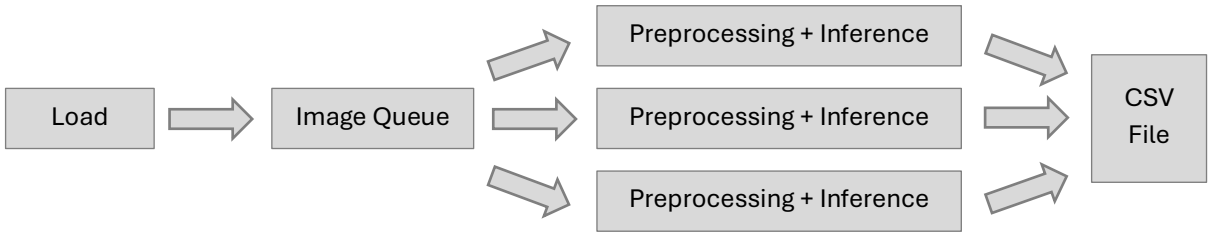
Background

An Optical Character Recognition (OCR) system is an application that extracts textual information from images. It has a wide range of uses in automation, document digitization, ID verification, license plate recognition, and other computer vision tasks. Modern OCR frameworks combine AI models with image processing techniques to achieve high accuracy even on noisy or low-quality inputs.

While these systems are effective, AI-based OCR can create performance bottlenecks as each image must be loaded, preprocessed, and passed to a model for inference. To improve efficiency, the sequential steps can be turned into a parallel pipeline that allows multiple images to be processed simultaneously.

Objective

Your task is to implement a multithreaded OCR pipeline that performs image preprocessing and inference in parallel. The main thread is responsible for loading and adding the images into a shared queue, which is then accessed by multiple worker threads that execute the necessary processing until the texts are extracted. The results are stored in a single CSV file (`result.csv`) containing the following details: ID (sequential integer), image filename, extracted text, and processing time (in milliseconds).

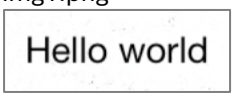


At the start of your program, the user should be prompted to input the directory of images to be processed. This will initiate the pipeline, which must run with *at least two worker threads*. You must employ the synchronization constructs and strategies discussed in class to ensure proper thread execution and safe access of shared resources. While you are allowed to use any built-in C++ synchronization construct, your solution *must include a binary or counting semaphore*. Once all images have been processed, the threads should terminate.

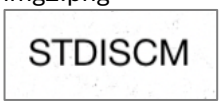
** Note: preprocessing and inference scripts will be provided*

Sample Images

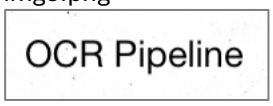
img1.png



img2.png



img3.png



Sample Result

ID	Image	Text	Processing Time
1	img1.png	Hello world	7 ms
2	img2.png	STDISCM	6 ms
3	img3.png	OCR Pipeline	8 ms

Report

Write a one or two-page PDF report that explains the following details

- Multithreading
- Synchronization
- Performance gain (use the speedup formula) and how this was achieved

$S = \frac{T_1}{T_n}$ T_1 total processing time with 1 thread

T_n total processing time with n threads

Submission

Via a Canvas online quiz